Communication Network Design - AY 2020/2021
Homework 1: Traffic sources

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1 Assignment

Using a packet sniffer (e.g., `Wireshark`), collect real-world traffic traces generated by a target network application at your choice as, e.g., `Zoom.us`, `Slack`, `Skype`, `Slack`, `Google Meet`, `Whatsapp`, ... Extract the packet generation (or reception) time and the packet size and analyze the resulting traffic profile. Propose reasonable stochastic models for packet inter-arrival time ($\{A_i\}$), batch size (marking, $\{B_i\}$), and packet size (workload, $\{W_i\}$). Use such models to generate a synthetic trace and compare it with the empirical data using proper representation tools (scatterplot, CDFs comparison, `qqplot`, `boxplot`, ...).

Write a short report (max 2 pages) with the result of your analysis and your considerations. Use the LaTex template available on moodle.